When should an image classifier refuse to answer?

Research question: improve accepted accuracy without hiding the coverage cost.

CIFAR-10 | all 10 classes | seed 42

10,000 train → 2,000 model validation → 1,500 calibration → 1,500 policy selection
Final test: 2,000 independent official test images, 200 per class

Frozen ResNet-18 + trained linear head

ImageNet weights; resize 256 / center crop 224; 30 head epochs.
Calibration T = 0.95617; raw / calibrated thresholds = 0.56 / 0.57.

A · Baseline

Raw; always predict

B · Rejection

Raw confidence ≥ 0.56

C · Calibration

Scaled; always predict

D · Combined

Scaled confidence ≥ 0.57

Clean rejection helps; corruption breaks reliability

All numbers below are final test. B/D show accepted accuracy and coverage (%).

Condition	Full accuracy	B acc. / cov.	D acc. / cov.
Clean	85.05%	90.85 / 88.00	90.90 / 87.90
Blur radius 1.5	22.55%	32.42 / 36.40	31.81 / 38.35
Noise std 0.12	16.20%	17.46 / 45.25	17.40 / 46.85
JPEG quality 15	44.95%	61.30 / 46.90	60.58 / 47.95

Clean D: +5.85 points accepted accuracy; 12.10% rejected.

Clean ECE: 0.0201 → 0.0187; NLL nearly unchanged. Labels stay the same.
Threshold sweeps trade coverage for risk; raw/scaled curves nearly overlap.
The clean 90% target is not a guarantee on corrupted images.

Live demo: confident errors and useful predictions rejected

Run the saved weights on the selected pixels. No fitting or test-based threshold changes.

Clean



clean/none



1. Confidently wrong and accepted
test:5806 | true: airplane | predicted: ship
Raw confidence 0.998; calibrated 0.999
D threshold 0.57; ACCEPT
Clean: ship / confidence 0.999

Clean



clean/none



2. Correct latent class, unnecessarily rejected
test:2998 | true: dog | predicted: dog
Raw confidence 0.550; calibrated 0.568
D threshold 0.57; REJECT
Clean: dog / confidence 0.568

Clean



jpeg/severe



3. Correct clean decision becomes a wrong accepted decision
test:4720 | true: ship | predicted: airplane
Raw confidence 0.988; calibrated 0.990
D threshold 0.57; ACCEPT
Clean: ship / confidence 0.857

What goes wrong?

Airplane → ship
Ambiguous silhouette/background; confidence ≈ 99.9%.

Correct dog → REJECT
56.84% is below frozen 57%.

Ship → airplane after JPEG
Lost details and blocking; still ≈ 99.0% confident.

Visual causes are hypotheses.
Fallback: saved case panel.
Reproducibility: repository README.